FEDERAL SPECIFICATION

INSULATION SLEEVING, THERMAL, (FLEXIBLE UNICELLULAR PIPE COVERING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers unicellular elastomeric thermal insulation sleeving for condensation control at temperatures -40° to +200°F. (40.0° to +93.3°C.).

1.2 Classification. The thermal insulation sleeving covered by this specification shall be tubular form, as specified (see 6.2).

1.3 The complete requirements for procuring thermal insulation sleeving described herein shall consist of this document and the latest issue of specification HH-I-1751/GEN.

2. APPLICABLE DOCUMENTS

2.1 Latest issue of Federal Specification HH-I-1751/GEN and documents referenced therein.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 355 - Test for Water Vapor Transmission of Thick Materials.
C 534 - Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form.
D 635 - Test for Flammability of Self-Supporting Plastic.
D 1056 - Specification for Sponge and Expanded Cellular Rubber Products.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Tubular.

3.1.1 Material. The material shall be unicellular composition, such as expanded synthetic elastomer, suitable for use as thermal insulation sleeving and condensation control, and unless otherwise specified, shall meet the requirements of ASTM C 534, except; for quality assurance, packing, marking and packaging requirements. The dimensions shall be as specified (see 6.2), and the tolerances shall be within the range specified in table I.

TABLE I. Dimensional tolerances, inches

	Tolerance
Inside diameter:	
Up to 3/8 incl	+1/16 -0
1/2 to 7/8 incl	+1/8 -0
1 to 1 1/2 incl	+3/16 -0
1 5/8 to 2 3/8 incl	+1/4 -0
Over 2 3/8	+3/8 -0
Wall thickness	
Up to 3/4	+1/8 -0
3/4 and over	+3/16 -0
Length, percent:	+3 -1

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:55Z
Check the source to verify that this is the current version before use.

3.1.2 Thermal conductivity. When tested in accordance with 4.4.1 of HH-I-1751/GEN the thermal conductivity (k) determined at a mean temperature of 75°F. shall not be greater than 0.30 Btu/Hr sq. ft. (F deg/in).

3.1.3 Fire resistance. When tested in accordance with 4.4.3 of this specification the fire resistance burning time shall be 8 seconds maximum and the char length shall be 3 inches, maximum.

3.1.4 Density. When tested in accordance with 4.4.4 of HH-I-1751/GEN the density rating shall be 4.5 to 8.5 pounds per cubic foot.

3.1.5 Flexibility. When tested in accordance with 4.4.4, of this specification, cracking, reupture, or permanent deformation shall constitute failure, of this test.

3.1.6 Water absorption. When tested in accordance with 4.4.1 of this specification, the water absorption shall be not greater than 10 percent by weight.

3.1.7 Water vapor permeability. When tested in accordance with 4.4.2 of this specification, the water vapor permaebility shall be not greater than 0.30 perm. inch.

3.2 Color. The color of the thermal insulation sleeving material shall be of the option of the contractor unless otherwise specified (see 6.2).

3.3 Workmanship. Insulation sleeving shall not have visual defects that adverely affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 The applicable Quality Assurance Provisions of Federal Specification HH-I-1751/GEN and the documents therein, shall apply with the following additional requirements:

4.2 Sampling.

4.2.1 In accordance with 4.2.1 of HH-I-1751/GEN, Unless otherwise specified, (see 6.2).

4.3 Examination.

4.3.1 End item. Insulation selected in accordance with 4.2.2 of HH-I-1751/GEN shall be examined for defects listed in table I of HH-I-1751/GEN.

4.3.2 Inspection of prepration for delivery. An inspection shall be made to determine that packing and marking requirements of the applicable contract or order are complied with. Defects shall be scored in accordance with table II of HH-I-1751/GEN.

4.4 Test methods. All tests shall be made on samples conditioned at 73.4° ± 3.6°F. (23° ± 2°C.) for 24 hours in an atmosphere of 50 ± 2 percent relative humidity. AQL for tests shall be 4.0 percent defective.

4.4.1 Water absorption. Determination of the water absorption shall be in accordance with ASTM D 1056 (see 3.1.6).

4.4.2 Water vapor permeability. Determination of the water vapor permeability shall be in accordance with ASTM C 355 (see 3.1.7).

4.4.3 Fire resistance. Determination of the fire resistance shall be in accordance with ASTM D 635 (see 3.1.3).

4.4.4 Flexibility. Determination of flexibility shall be in accordance with ASTM C 534 for type I tubular thermal insulation, (see 3.1.5).

4.4.5 Test specimens. Flat sheets from which the thermal insulation sleeving is produced, may be used as the test specimen.

5. PREPARATION FOR DELIVERY

5.1 In accordance with Federal Specification HH-I-1751/GEN and documents referenced therein:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Insulation units covered by this specification are intended for use to prevent excessive heat gain or loss, and to prevent condensation on plumbing, hot water heating, dual temperature air conditioning, and refrigeration pipe and equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Tubular inside diameter, wall thickness, and length (see 3.1.1).
(c) Size of lot, if different from 4.2.1 of HH-I-1751/GEN.
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2 of HH-I-1751/GEN.
(e) Marking required see 5.2 of HH-I-1751/GEN.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Army - CE, CG, SH

User activities:

Army - ME
Navy - YD, AS
Air Force - 84

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE - NBS
DOT - FIS
GSA-FSS, PCD, TCS
INTERIOR - BPA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.